

Table 1: A comparison of public cloud, private cloud and hybrid cloud usage

Feature	Public cloud	Private cloud	Hybrid cloud
Usage	Anyone can use public cloud services.	A single enterprise uses cloud services.	Cloud services are distributed among private and public clouds.
Data access	Virtualisation is used to build cloud services that are provided to the public.	A private cloud resides in the enterprise and behind a firewall, so only the user inside the enterprise can access and manage it.	Sensitive information is kept inside the private cloud, while other services can be hosted using a public cloud.
Data storage	Data of different enterprises is stored in a shared environment.	Data of a single enterprise is stored in a private environment.	Businesses can decide where to store data to secure it, making it cost-effective.
Data centre	Data centre is located anywhere on the internet.	The data centre location is inside the enterprise network.	Data centre is located inside the enterprises' network and on the internet.
Service management	Cloud service provider controls and manages the services.	Enterprises manage the services as they have their administrators.	Some services are managed by the enterprise, and the CSP manages some.
Security	Security is minimal.	High security.	High security.
Flexibility	Less flexible.	Less flexible.	Highly flexible.
Data recovery	Easily recovers enterprise data.	Data recovery is complicated.	Data is always secured.
Performance	Performance is low.	Improved performance.	Better performance.
Cost	Cost-effective	Fairly expensive	Cost-effective

organisations to design, build and deliver various business applications and network services. The latest version of Cloudify is 4.3, which introduced a totally new concept for container orchestration with Kubernetes (Kubernetes cloud native orchestration).

WSO2: This PaaS framework from the Apache community provides elastic scalability for any type of service, using the underlying infra cloud. WSO2 is a microservices-based architecture that supports various services, and fosters agility and flexibility with open source.

Apache Cloud Stack: This open source software platform is an infrastructure as a service (IaaS) cloud-computing platform. Various service providers use Cloud Stack to offer public cloud, private cloud and hybrid cloud services.

Eucalyptus: This open source software platform implements infrastructure as a service (IaaS) in a private or hybrid cloud computing environment. It provides quick and efficient computing services. It allows users to deliver their own data centres

into a private cloud and hence, extend the services to other organisations. Eucalyptus was designed to provide services compatible with Amazon's EC2 cloud and Simple Storage Service (S3).

Benefits of hybrid cloud adoption

Enhances business productivity: A hybrid cloud helps speed up business processes, supports collaboration, and provides cost-effective solutions to free up IT budgets for innovative, revenue-generating projects.

Reduces costs: It helps in reducing operating and capital costs, and improves performance, productivity and business agility by offering a flexible, scalable solution.

Flexibility: Enterprises can choose the optimal environment for each workload based on performance, security, and cost, and can easily migrate their workloads to and from different cloud models whenever necessary.

Cost optimisation: A hybrid cloud model allows users to choose from the environment where they

want to run their workloads most cost-effectively. By moving non-critical workloads to the public cloud, enterprises can reduce their on-premises infrastructure costs and pay only for the resources they consume.

Reliability: In this model, if one cloud goes down, some functionality will still be available to users from the other deployed clouds. Generally, one public cloud can be used as a backup to another cloud.

Resilience: In a hybrid cloud model, an enterprise can run applications redundantly in both environments, i.e., public and private clouds. Thus, components of a single workload can interoperate and run in both environments.

Risk management: A hybrid cloud helps in mitigating risks with a single, unified, cybersecurity solution.

Managing legacy systems: A hybrid cloud model can bridge the gap between legacy and new systems, helping save costs.

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